Punto fijo

$$f(x) = -x^2 + 1.8x + 2.5 \qquad x_0 = 5$$

$$g(x) = \sqrt{1.8x + 2.5}$$

$$x_1 = g(x_0) = \sqrt{1.8(5) + 2.5} = 3.391165$$

$$e_a = \left| \frac{3.391165 - 5}{3.391165} \right| \times 100 \approx 47.44\%$$

$$x_2 = g(3.391165) = \sqrt{1.8(3.391165) + 2.5}$$
$$= 2.933274$$

$$e_a = \left| \frac{2.933274 - 3.391165}{2.933274} \right| \times 100 = 15.61\%$$

$$x_3 = g(2.933274) = \sqrt{(1.8)(2.933274) + 2.5}$$
$$= 2.789246$$

$$e_a = \left| \frac{2.789246 - 2.933274}{2.789246} \right| \times 100 = 5.16\%$$

$$x_4 = g(2.789246) = \sqrt{(1.8)(2.789246) + 2.5}$$
$$= 2.742379$$

$$e_a = \left| \frac{2.742374 - 2.789246}{2.742379} \right| \times 100 = 1.709\%$$

$$x_5 = g(2.742374) = \sqrt{(1.8)(2.742379) + 2.5}$$
$$= 2.726955$$

$$e_a = \left| \frac{2.726955 - 2.742379}{2.726955} \right| \times 100 = 0.565\%$$

$$X_6 = g(2.726955) = \sqrt{1.8(2.726955) + 2.5} =$$
$$= 2.718183$$

$$e_a = \left| \frac{2.718183 - 2.726955}{2.718183} \right| \times 100\% = 0.187\%$$

$$X_7 = g(2.718183) = \sqrt{1.8(2.718183) + 2.5}$$
$$= 2.720056$$

$$e_a = \left| \frac{2.720056 - 2.718183}{2.720056} \right| \times 100\% = 0.061\%$$

$$X_8 = g(2.720056) = \sqrt{1.8(2.720056) + 2.5}$$
$$= 2.719639$$

$$e_a = \left| \frac{2.719639 - 2.720056}{2.719639} \right| \times 100\% = 0.015\%$$

Comprobación

$$f(2.719639) = -(2.719639)^2 + 1.8(2.719639) + 2.5$$
$$= -0.0010 \Rightarrow \text{cerca de } 0$$

Newton Raphson

$F'(x) = -2x + 1.8$

$X_1 = F(5) = -(5)^2 + 1.8(5) + 2.5 = -13.5$

$F'(5) = -2(5) + 1.8 = -8.2$

$X_1 = -\left(\frac{-13.5}{-8.2}\right) + 5 = 3.353659$

$e_a = \left| \frac{3.353659 - 5}{3.353659} \right| \times 100 = 49.09\%$

$X_2 = F(3.353659) = -(3.353659)^2 + 1.8(3.353659) + 2.5$
$= -2.710738$

$F'(3.353659) = -2(3.353659) + 1.8 = -4.907318$

$X_2 = 3.353659 - \left(\frac{-2.710738}{-4.907318}\right) = 2.801252$

$e_a = \left| \frac{2.801252 - 3.353659}{2.801252} \right| \times 100 = 19.72\%$

$X_3 = F(2.801252) = -(2.801252)^2 + 1.8(2.801252) + 2.5$
$= -0.303258$

$F'(2.801252) = -2(2.801252) + 1.8 = -3.802504$

$X_3 = 2.801252 - \left(\frac{-0.303258}{-3.802504}\right) = 2.721500$

$e_a = \left| \frac{2.721500 - 2.801252}{2.721500} \right| \times 100\% = 2.930\%$

$$X_4 = F(2.721500) = -(2.721500)^2 + 1.8(2.721500) + 2.5$$
$$= -0.007137$$

$$F'(2.721500) = -2(2.721500) + 1.8 = -3.643000$$

$$X_4 = 2.721500 - \left(\frac{-0.007137}{-3.643000}\right) = 2.719541$$

$$e_a = \left|\frac{2.719541 - 2.721500}{2.719541}\right| \times 100$$

$$e_a = 0.072\%$$

$$X_5 = F(2.719541) = -(2.719541)^2 + 1.8(2.719541) + 2.5$$
$$= -0.000004$$

$$F'(2.719541) = -2(2.719541) + 1.8 = -3.639082$$

$$X_5 = 2.719541 - \left(\frac{-0.000004}{-3.639082}\right) = 2.719540$$

$$e_a = \left|\frac{2.719540 - 2.719541}{2.719540}\right| \times 100 = 0.000003\%$$

Comprobación

$$F(2.719540) = -(2.719540)^2 + 1.8(2.719540) + 2.5$$
$$= -0.000725$$